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**OrgVault: Repository of Campus Student Organization Narrative and Summary
Reports for the Office of the Social Development and Welfare at CSU-Aparri.**

A Capstone Project Proposal
Presented to the Faculty of the
College of Information and Computing Sciences
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Aparri Campus

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Chapter 1

The Problem and Its Background

Every educational institution recognizes the importance of student organization activities in promoting leadership, collaboration, and active student engagement through academic, cultural, social, and extracurricular programs. These activities provide students with opportunities to develop communication, organizational, and decision-making skills beyond the classroom. Moreover, student organizations contribute significantly to campus culture, student development, and institutional growth.

However, the effective management and monitoring of student organization narrative reports, accomplishment reports, and summary reports require accurate documentation and systematic tracking. Proper record-keeping ensures transparency, accountability, and institutional support. Without a reliable system, monitoring submissions, reviewing reports, and retrieving past records can become inefficient and time-consuming.

Despite the increasing number of student organizations and campus activities, many institutions still rely on manual or semi-manual methods such as printed documents, logbooks, spreadsheets, and email submissions to manage narrative, accomplishment, and summary reports. These traditional approaches present several limitations. Paper-based records require physical storage space and are vulnerable to damage, loss, or misplacement. Retrieving previous reports can be difficult, especially when documents are not organized in a centralized and searchable system.

The rapid advancement of information technology has significantly transformed administrative processes in higher education institutions. One of the areas affected by digital transformation is the monitoring and documentation of student organization activities. Traditional paper-based reporting



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systems often result in delayed submissions, misplaced documents, and inefficient monitoring processes.

Jones and Taylor (2022) examined the impact of real-time monitoring of student activities on academic performance. Their study emphasized the role of systems that track both academic and extracurricular activities in enhancing student development. This research is relevant to the present study as it supports the use of digital platforms to monitor and evaluate student participation, which can improve accountability, engagement, and overall performance in organizational activities.

In the Philippine setting, Delos Santos (2020) stressed the importance of digitizing student organization documentation to preserve institutional records and support decision-making. The study revealed that digital reporting systems enhance accessibility and prevent data loss, especially in universities with multiple recognized organizations. This supports the need for a web-based monitoring system specifically designed to handle narrative, accomplishment, and summary reports of campus student organizations.

In addition, updating or revising activity records often requires rewriting or duplicating documents, increasing the risk of inconsistencies and errors. Security and data integrity are also major concerns in manual record-keeping systems. Physical documents can easily be misplaced, stolen, or destroyed, resulting in permanent loss of information if no backup exists. Unlike digital systems, manual records lack reliable mechanisms for access control, data backup, and recovery. These limitations hinder administrators and organization advisers from efficiently monitoring activity status, participation, and compliance with institutional policies.

With the advancement of web-based technologies, institutions are increasingly adopting digital platforms to improve information management and accessibility. A web-based system allows data to be securely stored, easily updated, and quickly retrieved anytime and anywhere. Such platforms



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support centralized management, real-time monitoring, and automated reporting, making them more efficient and reliable than traditional paper-based methods.

This project is entitled "CSORMS: Campus Student Organization Narrative and Summary Reports for the Office of the Social Development and Welfare at CSU-Aparri." The proposed system provides an online platform for recording, organizing, and managing student organization activities within the campus. The system is designed to allow student organization officers to submit activity details, while administrators and advisers can review, monitor, and generate reports as needed. Its primary purpose is to improve documentation, enhance transparency, and ensure efficient management of student organization activities while maintaining secure, accessible, and long-term digital records for institutional use.

The proposed project "Campus Organization Report Management System" supports sustainable development by enhancing educational quality, encouraging technological innovation, and strengthening institutional transparency. By integrating digital solutions into campus management processes, the study contributes to the achievement of SDGs 4, 9, and 16 through improved student engagement, efficient administration, and accountable organizational governance.



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Conceptual Framework

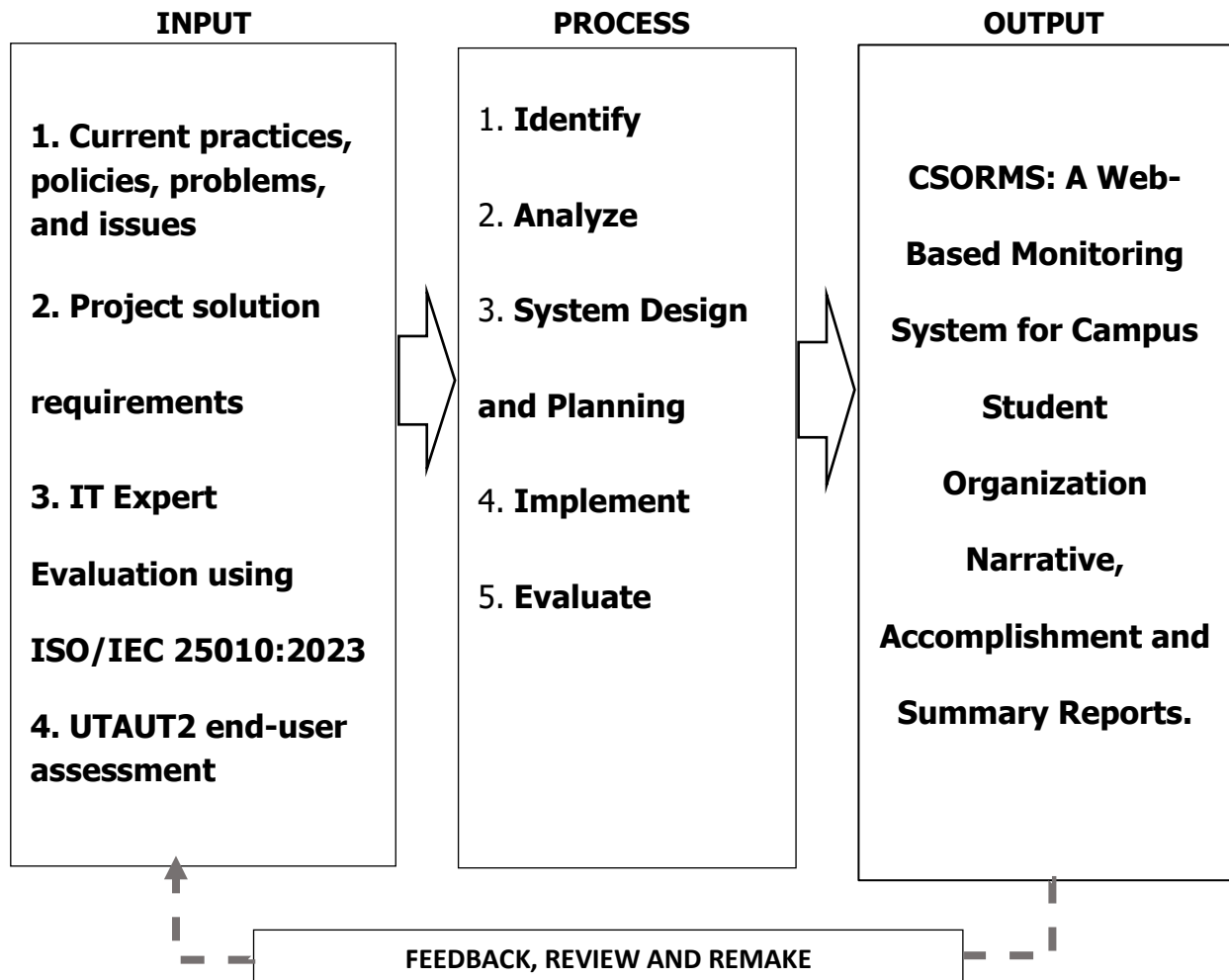


Figure 1: Conceptual Framework of the developed Project

The figure below shows the three (3) stages of our proposed project concept. The Input, shows the starting point of the proposed project where all information needed by the researchers is gathered in order to have better understanding of what is needed to achieve the proposed project objectives. The Process, this shows the different levels that the researchers would undergo, where in these includes (1) Identifying, identification of the data, records, policies, SOP's forms, and reports used and implemented by the target agency is important to start the project requirement



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analysis. (2) Analyze, where all information is being describe and interpreted to come up with a specific plan of action for the design and development of the project (3) System Design and Planning, this stage, the researchers is to create the actual project based to what is planned at the Analysis phase. This includes wire framing and actual coding. (4) Implement, this stage would require the researchers to test the proposed project "system" to its actual performance with the users/intended users of the project to experience of the project's relevance and effectiveness.

Objectives of the Study

The proposed project **"CSORMS: A Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment and Summary Reports."** aims to address the challenges encountered in documenting, monitoring, and managing student organization reports within the campus. The current system is primarily manual or semi-manual, relying on paper-based records and scattered digital tools, which are time-consuming, inconsistent, and prone to errors. These issues result in delayed approvals, difficulty in tracking activity progress, and limited monitoring capability.

Specifically, the study seeks to answer the following questions:

1. What are the current practices, policies, problems, and issues encountered in managing student organization activities using the existing manual or semi-manual system?
2. What web-based project solution can be developed to address the identified problems of the current system?
3. To what extent does the developed CSUA-CSORMS system comply with the ISO/IEC 25010:2023 software quality standards in terms of:
 - a. Functional Suitability



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b. Performance Efficiency

c. Compatibility

d. Usability

e. Reliability

f. Security

g. Maintainability

h. Portability

4. What is the level of user acceptance of the developed system based on the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) in terms of:

a. Performance Expectancy

b. Effort Expectancy

c. Social Influence

d. Facilitating Conditions

e. Hedonic Motivation

f. Price Value (if applicable)

g. Habit

Scope and Limitations

This proposed study entitled Design and Development of a Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment, and Summary Reports focuses on the design and development of a web-based platform intended to monitor, manage, and centralize the submission of narrative and accomplishment reports within the campus student organizations. The system covers report submission, monitoring of report status, approval and validation of reports by



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authorized administrators, and viewing of uploaded narrative, accomplishment, and summarized documents.

The study is limited to web-based implementation and does not include mobile application or offline software development and specifically for campus student organizations within the Cagayan State University-Aparri Campus and may not be applicable to other universities without modification. It focuses solely on the monitoring and management of organization reports. The study will utilize relevant organizational report data provided by the campus student organizations and institutional administrators, provided that proper authorization and consent are secured. These data will be used exclusively for system development, testing, and evaluation purposes.

The scope of the project is limited to providing a centralized web-based monitoring system that allows student organizations to submit narrative, accomplishment, and summarized reports through digital forms. Organization officers can upload report documents, while administrators can review, monitor submission status, validate entries, and access stored report records. The system will be accessible using desktop computers, laptops, and mobile devices with stable internet connectivity.

Significance of the study

The Office of Student Development and Welfare, the system offers organized and reliable activity records that support reporting, evaluation, and decision-making.

To Staff of Office of Student Development and Welfare, for administrative staff, the system simplifies the processes of reviewing, verifying, and monitoring submitted organizational activities.

The Student Organization Officers, the system provides a centralized and efficient platform for submitting, managing, and tracking student organization activities.



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To Organization Advisers, the system enables advisers to monitor, review, and approve activities in compliance with institutional policies.

To the CSU-Aparri, the system promotes transparency, accountability, and digital transformation in managing student organization activities.

The Researchers, would serve as their knowledge and create solutions and implement system for the campus.

To the Future Researchers, the system will provide a guide of references for relevant ideas, knowledge, and information for those who may experience related system.

Definition of terms

Accomplishment Report – A formal document submitted by a student organization detailing the outcomes, achievements, and results of a conducted activity or program.

Administrator – Authorized personnel from the Office of Student Development and Welfare responsible for reviewing, validating, and monitoring submitted organization reports within the system.

Campus Student Organization Report Management System (CSORMS) – The proposed web-based monitoring system designed to manage, record, organize, and monitor narrative, accomplishment, and summary reports of campus student organizations.

Functional Suitability – A software quality characteristic under ISO/IEC 25010 that evaluates whether the system provides functions that meet stated and implied needs.

Hedonic Motivation – A construct under Unified Theory of Acceptance and Use of Technology 2 referring to the enjoyment or pleasure derived from using the system.



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Maintainability – A software quality attribute under ISO/IEC 25010 that measures the ease with which the system can be modified, updated, or improved.

Narrative Report – A detailed written account describing the planning, implementation, and overall conduct of a student organization activity.

Student Organization Officer – A recognized student leader responsible for submitting activity reports and managing organization records within the system.

Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) – A technology acceptance model developed by Venkatesh et al. that explains user intentions and behavior in adopting information systems.



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Chapter 2

Review of Related Literature and Studies

Foreign Literatures and Studies

The continuous advancement of information technology has transformed how educational institutions manage student organizations, activities, and administrative processes. Universities increasingly adopt digital systems to replace manual operations in order to improve efficiency, coordination, and accessibility of organizational data. Blancaflor et al. (2021) developed the *Cardinal Connect Event Management System*, a web-based platform designed to assist student organizations in managing event applications, implementation, and post-activity evaluation. The study highlighted that centralized systems reduce administrative workload and improve coordination between student organizations and university offices, particularly during the transition to online environments.

Higher education institutions also face challenges in managing large volumes of organizational documents. Arruejo and Arruejo (2024) proposed a Student Organizations Management System integrated with descriptive analytics and dashboards to support real-time monitoring and decision-making. Their findings revealed that automation and visualization tools enable administrators to analyze organizational performance efficiently while optimizing institutional resources.

To address management inefficiencies, Aziz et al. (2021) introduced an Integrated Student Activities Management System that centralized processes such as proposal submission, budget approval, activity monitoring, and reporting. The study found that digital platforms eliminate delays caused by manual documentation and improve monitoring efficiency. Similarly, Shara and Sarala (2025) developed an Online Event Management System integrating registration, documentation uploads, and event archiving features. The system enhanced accessibility while minimizing manual workloads in organizing academic events.



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Advancements in monitoring technologies further support institutional management systems. Honma and Kiyohara (2023) proposed activity-tracking mechanisms capable of monitoring student engagement across campus environments using connected digital technologies. Their research demonstrated that tracking systems help institutions identify students needing academic support and improve overall performance outcomes.

Human-computer interaction and smart campus technologies also contribute to effective student management. Wang (2022) explored data acquisition and monitoring systems that enhance organizational management through automated interaction and evaluation processes. Likewise, Li et al. (2021) developed a smart campus management system utilizing wearable devices and data analytics to monitor student participation and behavioral patterns for accurate institutional decision-making.

Digital tools designed for educational activity management have likewise improved operational efficiency. Sarasa-Cabezuelo and Caballé (2020) created a system for managing seminars and workshops that facilitated participant monitoring, feedback collection, and notification dissemination. Their findings indicated that automated systems significantly improve communication and coordination within academic environments.

Mobile and cloud-based technologies further strengthen modern monitoring systems. Zhang and Wu (2021) highlighted that mobile applications enable real-time monitoring of student participation and engagement through accessible platforms. Meanwhile, Lee and Lee (2022) emphasized that cloud-based systems provide scalable storage, automated reporting, and seamless communication among stakeholders, making them suitable for large educational institutions.

Alshibly (2019) explained that the success of information systems depends on system quality, information quality, and user satisfaction. The study emphasized that well-designed systems



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improve efficiency and user performance. It also highlighted that organizations benefit from systems that are reliable and easy to use. Moreover, user trust plays a significant role in system adoption. This supports the need for a functional and user-friendly reporting system.

Dennis et al. (2020) discussed that systems analysis and design are essential in developing effective information systems. They emphasized understanding user requirements before system development. The study also explained the importance of structured processes in system creation. Proper system design leads to improved usability and efficiency. This is relevant in designing a reporting management system for student organizations.

Laudon and Jane P. Laudon (2021) stated that information systems support organizational operations and decision-making. They highlighted that digital systems improve data management and communication. The authors emphasized the importance of centralized systems in handling large volumes of information. They also noted that organizations rely on information systems for efficiency and productivity. This is applicable to managing student organization reports.

Marston et al. (2020) explained that cloud computing enhances data storage and accessibility. The study emphasized that cloud-based systems allow users to access data anytime and anywhere. It also discussed improved scalability and flexibility in cloud environments. Organizations benefit from reduced infrastructure costs. This supports the use of online systems for report submission.

O'Brien and Marakas (2020) stated that management information systems improve organizational performance. They emphasized that systems help in collecting, processing, and storing data efficiently. The study also highlighted improved communication through digital platforms. It explained that systems support decision-making processes. This is relevant in managing and monitoring reports. Roger S. Pressman and Maxim (2020) emphasized that software



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engineering ensures system quality and reliability. They discussed that proper development processes reduce system errors.

The study also highlighted the importance of testing and maintenance. Well-developed systems improve user experience. This supports the development of a reliable reporting system. Sommerville (2020) explained that software systems must be designed with user needs in mind. The study emphasized system reliability, efficiency, and usability. It also discussed the importance of system maintenance. Proper system design ensures long-term effectiveness. This is relevant in building a sustainable reporting system. Stair and Reynolds (2021) stated that information systems are essential in modern organizations. They help improve productivity and data management. The study emphasized accurate and timely information. It also highlighted the importance of system security. This supports the implementation of secure reporting systems.

Turban et al. (2021) discussed digital transformation in organizations. They emphasized that information systems enhance operational efficiency. The study highlighted automation and data-driven decision-making. It also explained improved organizational performance. This supports the use of automated reporting systems. Alam and Noor (2020) emphasized the role of system quality in user satisfaction. They explained that reliable systems improve trust and usage. The study also highlighted the importance of service quality. Better systems lead to higher user engagement. These supports developing a user-friendly system.

Rana et al. (2021) discussed the adoption of e-government systems. They emphasized transparency and accessibility in digital platforms. The study highlighted factors affecting system acceptance. It also explained the importance of usability. This is relevant to system adoption in academic institutions.



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Sarker (2021) explained that data-driven systems improve decision-making. The study emphasized the importance of data analytics. It highlighted the role of digital systems in managing large data. Organizations benefit from organized data processing. This supports structured reporting systems. Al-Fraihat et al. (2020) evaluated e-learning systems and identified key success factors. They emphasized system quality, information quality, and user satisfaction. The study highlighted the importance of usability. It also discussed system effectiveness in digital environments. This is applicable to reporting systems. Kumar and Singh (2022) explained that web-based systems improve organizational efficiency. They emphasized accessibility and convenience. The study highlighted reduced manual work. It also discussed improved communication. This supports online reporting platforms.

Local Literatures and Studies

In the Philippine educational context, the implementation of digital management systems has become increasingly important in addressing challenges related to student organization administration and documentation. Marzan et al. (2025) developed a Club and Academic Organization Management System with decision-support features that centralized organizational data and automated reporting processes. The study revealed that system implementation reduced administrative workload while improving accountability and decision-making among organization officers.

Manual documentation remains a common issue among local institutions, often resulting in delayed reporting and incomplete records. An information systems study conducted at Satya Wacana Christian University (2024) emphasized that digital tracking platforms are necessary to address inefficiencies caused by traditional record-keeping practices in student organizations.



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Technological solutions for monitoring student participation have also been explored locally. Agustin (2021) introduced an RFID-based monitoring system designed to enhance attendance tracking and participation verification in educational institutions. The system improved accuracy, minimized human error, and enabled administrators to analyze participation trends effectively. Supporting this approach, RFID-based attendance monitoring systems with SMS notification features were shown to provide real-time reporting and data-driven decision-making capabilities for managing campus activities.

Furthermore, studies examining student organizational involvement highlight the importance of structured monitoring systems. Comighud et al. (2025) found that active participation in campus organizations promotes leadership development, time management skills, and emotional resilience among students. However, challenges such as workload balance and organizational stress indicate the need for institutional systems capable of monitoring participation and providing administrative support.

Additionally, college management digitalization initiatives discussed by Manohar et al. (2025) emphasized that transforming manual institutional workflows into intelligent applications enhances transparency, streamlines operations, and promotes efficient communication among stakeholders. These findings support the adoption of automated systems for managing student organization activities within higher education institutions.

Overall, local studies demonstrate that digital platforms and monitoring systems significantly improve organizational management, participation tracking, and administrative efficiency in Philippine universities. The adoption of centralized systems enables institutions to manage records effectively while supporting student engagement and informed decision-making processes.



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Bolasco, Costales, Saldivar, Valdez, and Cruz (2025) designed *LabTrack*, a system developed to monitor student attendance and computer laboratory usage through an automated tracking mechanism. The study utilized usability surveys to evaluate system effectiveness and revealed that automation significantly reduced recording errors, improved data accuracy, and enhanced user engagement compared to traditional manual attendance monitoring. Although the system primarily focused on laboratory utilization, its tracking principles are applicable to monitoring participation in student organization activities. Furthermore, the system enabled administrators to generate reports and analyze participation trends efficiently, demonstrating its usefulness in educational management environments. This study supports the development of reliable automated tracking systems adaptable to organizational monitoring functions.

A group project conducted at Iloilo Science and Technology University (2023) implemented a Student Organization Management System designed for membership management, event monitoring, and organizational record keeping. The system automated attendance recording and activity tracking while providing centralized storage for organizational documents and reports. Results indicated improved communication among organization officers and reduced administrative workload. Although not peer-reviewed research, the project demonstrates the practical application of information systems within the Philippine educational context and highlights the importance of institution-specific solutions tailored to school policies and student organizational needs.

Similarly, Arruejo and Arruejo (2020) developed an Adaptive Decision-Support System with Descriptive Analytics for student organizations at the University of Northern Philippines. The study emphasized the transition from basic data storage toward intelligent data analysis through dashboard visualization of Key Performance Indicators (KPIs). The system enabled administrators to monitor organizational performance and compliance in real time, improving institutional decision-



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making processes. The research demonstrates how analytics-driven platforms enhance monitoring efficiency and organizational governance within higher education institutions.

Navarra and Antonio (2025) emphasized that accurate and real-time information tracking systems have become essential tools for educational institutions in monitoring student attendance, which directly contributes to academic performance and student safety. Their study implemented an online monitoring and data tracking system at Isulan National High School to replace traditional paper-based attendance recording methods that often-prevented parents from being informed about student absenteeism. The system incorporated Short Message Service (SMS) notifications to enable real-time communication between the school and guardians. Findings revealed that approximately 95% of stakeholders considered the technology-driven monitoring approach effective in improving transparency, communication, and overall educational management. The study supports the adoption of automated monitoring systems capable of enhancing institutional oversight and real-time participation tracking applicable to student organization activities.

A study conducted at **Quirino State University** (2025) investigated the use of QR code recognition for monitoring attendance during school activities. Traditional attendance methods were found to be time-consuming and susceptible to student dishonesty, such as signing in for others or leaving early. Using the **Technology Acceptance Model (TAM)**, researchers found that student organization officers perceived QR-based attendance systems as highly **useful** for maintaining accurate records and simplifying attendance summaries. By equipping student leaders with the tools to monitor peers effectively, these systems promoted a **transparent smart campus environment**. The study concluded that high perceived **usefulness** and **ease of use** strengthened student leaders' commitment to adopting these automated attendance systems for institutional events.



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According to Mallillin, Caranguian, and Mallillin (2023), involvement in student organizations serves as an administration's conduit for delivering essential services to the student body. Their study highlights that co-curricular participation provides students with the power of expression and teaches critical time management skills necessary for balancing organizational duties with academic performance. In the Philippine context, the readiness and implementation of these organizational management systems are key criteria used by accrediting bodies to evaluate institutional quality. Effective tracking of these activities thus supports the theory of learning transformation and institutional sustainability.

Most Philippine educational institutions still rely on manual paper-based approaches for attendance tracking, highlighting a critical need for efficient digital solutions according to Acasamoso and Avila (2021), most. The implementation of Radio Frequency Identification (RFID) based systems with SMS notifications allows for the automatic recording of student time-in and time-out data. This technology eliminates the inefficiencies of pen-and-paper processes and provides real-time awareness for parents regarding their child's whereabouts. Additionally, the integration of data analytics within these systems helps administrators evaluate attendance trends and optimize institutional resource management.

According to Western Mindanao State University (2025), the manual coordination of professional development programs often suffers from fragmented workflows, delayed communication, and inconsistent data handling. To address these limitations, a web-based Centralized Event Management System (CEMS) was developed to integrate registration, scheduling, and QR code-based attendance tracking into a single responsive platform. The study utilized the Agile development model and ISO 9126 software quality metrics to ensure the system's effectiveness and reliability, resulting in high user satisfaction. This digital solution provides a secure



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and automated environment for managing training records and digital certifications, thereby increasing transparency and accessibility for both faculty and students.

According to researchers at Pangasinan State University (2025), traditional paper-based coordination for student councils often leads to scheduling conflicts, delays, and poor information sharing. The development of CouncilSync, a digital event management platform, addresses these inefficiencies by offering centralized calendars, real-time email notifications, and archival features. The system integrates event feeds and media uploads to engage the student body more effectively while reducing the administrative burden on council officers. Evaluation shows that the platform significantly improves collaboration and execution accuracy, resulting in reduced scheduling conflicts and higher attendance rates for various student-led activities.

According to the study of Ugalingan and Antonio 2025, modernizing administrative processes through a web-based school record management system enhances institutional efficiency, accuracy, and reliability. This digital solution addresses traditional issues like data retrieval delays and human errors, which significantly improves the speed of processing student records. By integrating such systems into school operations, institutions can reduce administrative workloads and support better decision-making through streamlined data management. Ugalingan & Antonio, 2025.

Garcia and Revano (2020) developed a web-based student information system. They found that digital systems improve data management. The study emphasized faster transactions and better accessibility. It also highlighted improved communication. This supports student-related systems. Dela Cruz and Torres (2021) developed a document tracking system for academic institutions. They found that digital systems reduce lost documents. The study emphasized improved tracking and monitoring. It also highlighted efficiency in document handling. This supports report management systems.



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Santos and Mendoza (2022) implemented an automated reporting system. They found improved efficiency and reduced delays. The study emphasized system automation. It also highlighted improved monitoring of submissions. This supports CORMS. Villanueva and Ramos (2023) developed a web-based management system for student activities. They found improved monitoring and coordination. The study emphasized centralized systems. It also highlighted better organization of data. This supports student organization systems.

Reyes and Bautista (2024) discussed digital transformation in academic institutions. They emphasized the importance of modern systems. The study highlighted improved efficiency and performance. It also discussed system integration. These supports adopting digital reporting systems. Torres and Aquino (2023) studied cloud-based document management systems. They found improved storage and accessibility. The study emphasized data security and reliability. It also highlighted system scalability. This supports cloud-based reporting systems.



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Chapter 3

Methods and Procedures

Research Design

The study employed a Descriptive–Developmental Research Design. The descriptive component focused on identifying the current practices, policies, problems, and issues encountered in managing narrative, accomplishment, and summary reports of campus student organizations. This involved gathering information regarding the existing manual or semi-manual processes implemented by the Office of Student Development and Welfare.

The developmental component focused on the design, development, and implementation of the proposed web-based monitoring system. The developed system was evaluated based on the software quality characteristics defined under ISO/IEC 25010. Furthermore, user acceptance of the system was assessed using the constructs of Unified Theory of Acceptance and Use of Technology 2 to determine the level of acceptance among intended users.

This research design ensured that the system was both technically compliant and acceptable to its users.

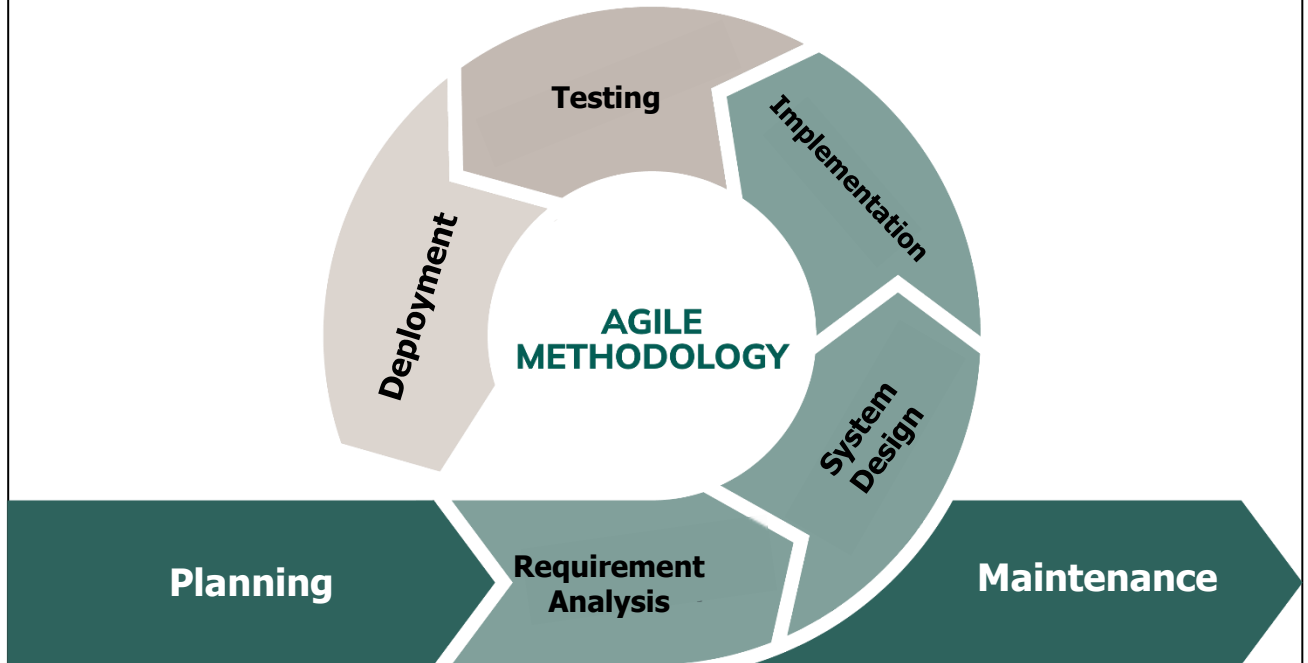
Software Development Methodology

The study utilized the Agile Sprint Software Development Model in designing and developing the proposed system. Agile Sprint methodology emphasizes iterative development, collaboration, flexibility, and continuous improvement. The system was developed in incremental cycles to allow feedback and refinements throughout the development process.



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Figure 2. Agile Sprint Software Development Model



The **Agile Development Model**, includes 7 stages listed as follows:

Stage 1 – Planning Phase

The planning phase focused on establishing the project scope, objectives, and required resources for the development of CSORMS: A Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment and Summary Reports. During this phase, Key stakeholders such as student organization officers, organization advisers, the Office of Student Development and Welfare, and the technical development team. Their roles and responsibilities were determined to ensure proper coordination throughout the project lifecycle.

Stage 2 – Requirement Analysis Phase

In this phase, the proponents gathered and analyzed information regarding the features and functionalities required for the development of the CORMS: A Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment and Summary Reports. The proponents conducted observations and interviews with research advisers, student organization officers, and



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administrative staff from the Office of Student Development and Welfare to determine their experiences and challenges with the existing manual or semi-manual reporting system.

The information collected from the participants served as the foundation in identifying system requirements, necessary features, user access levels, and operational specifications. These requirements were carefully documented to guide the system design and development process.

For the overall structure of the proposed project, the team utilized a web-based system architecture, as shown in Figure 3.

Figure 3. System architecture of the proposed Project

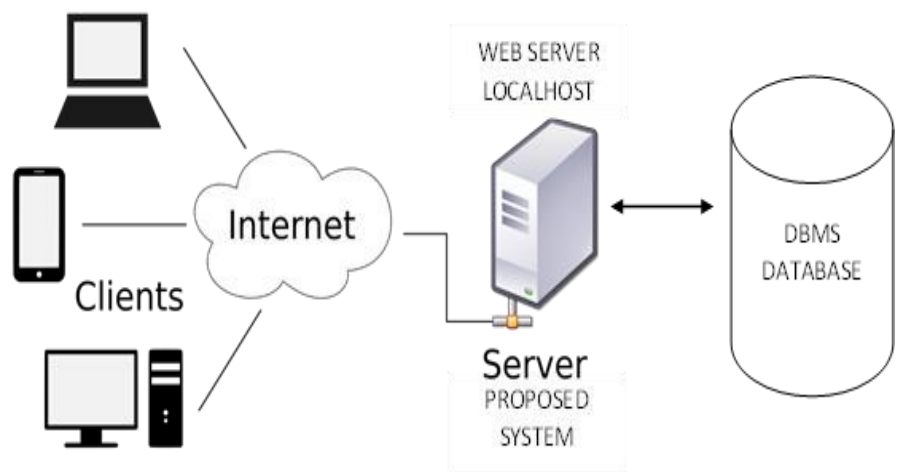


Figure 3 illustrates the system architecture of the proposed project. The system resides on a web server and is accessible through desktop computers, laptops, and smartphones via internet connection or localhost during testing. All submitted data and records are stored in a centralized database managed by the web server through a database management system. The architecture ensures secure storage, centralized access, and real-time monitoring of student organization reports.



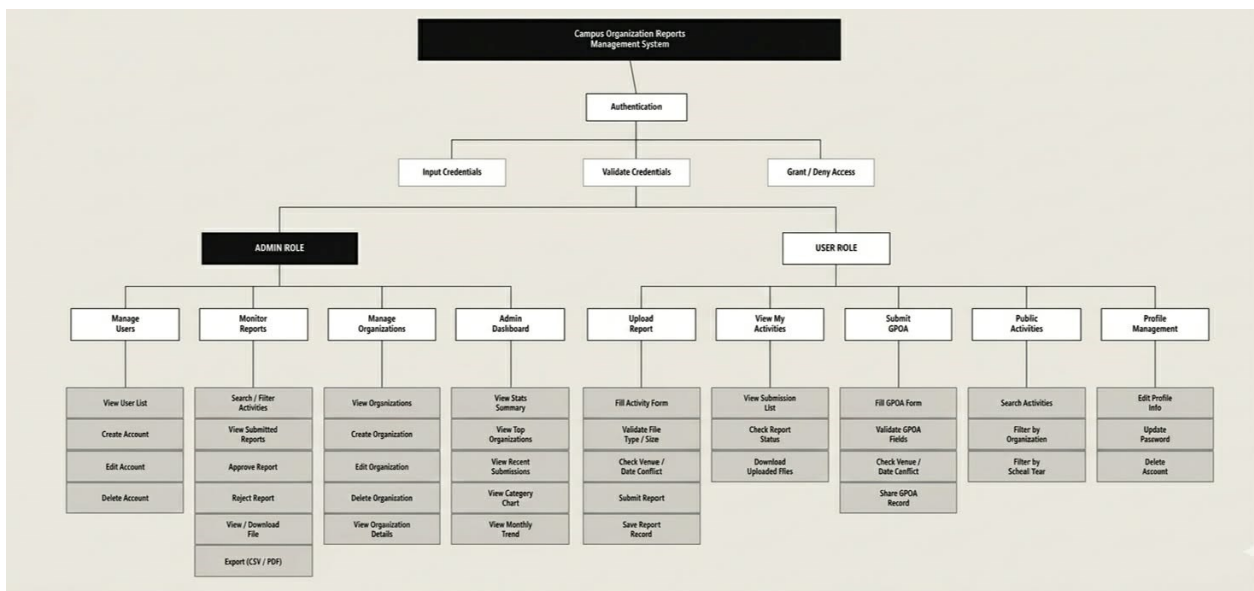
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Stage 3 – System Design Phase

After determining the system requirements, the proponents proceeded to the design phase. This stage focused on designing the system's user interface (UI), database structure, and overall workflow.

The user interface was designed to be intuitive and user-friendly for student organization officers, advisers, and administrators. The system includes modules for report submission, monitoring of submission status, validation, approval, and report viewing.

Figure 4. Hierarchical Input-Process-Output of the System



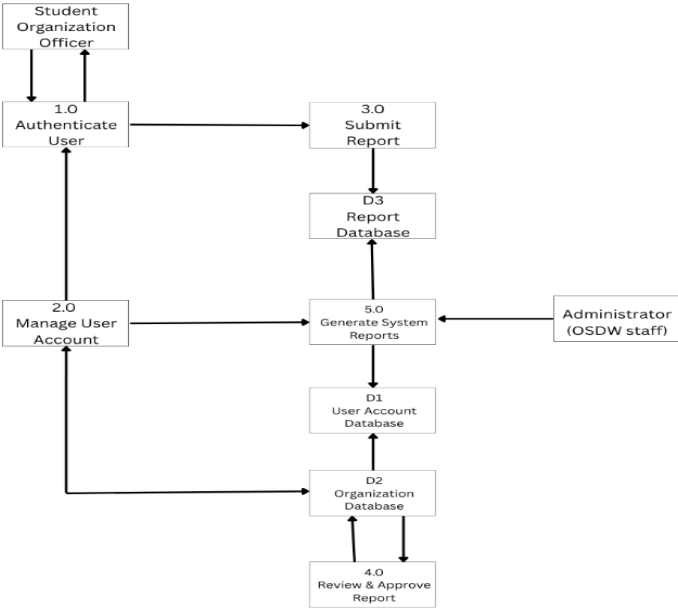
The Hierarchical Input-Process-Output (HIPO) diagram presents the overall functional structure of the Campus Student Organization Reports Management System (CSORMS). The system begins with the Authentication module, which is responsible for validating user credentials and determining whether access will be granted or denied. Upon successful authentication, the system divides into two primary role-based components: the Admin Role and the User Role. Under the Admin Role, three major modules are provided. The Manage Users module allows the administrator



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to create, edit, and delete user accounts, as account registration is exclusively controlled by the administrator. The Monitor Reports module enables the administrator to search and filter submitted reports, review uploaded activities, approve or reject submissions, and export report data in CSV or PDF format. The Admin Dashboard module displays a summary of system data, including total submissions, top-performing organizations, and recent activity reports. Under the User Role, the system also provides three key modules. The Upload Report module allows users to submit activity reports by filling out a form, validating file type and size, checking for schedule conflicts, and saving the report to the database. The View My Activities module enables users to track their submitted reports, monitor their status, and download uploaded files. The Submit GPOA module allows users to encode and store their General Plan of Activities.

Figure 5. Data Flow Diagram level 1 of the System





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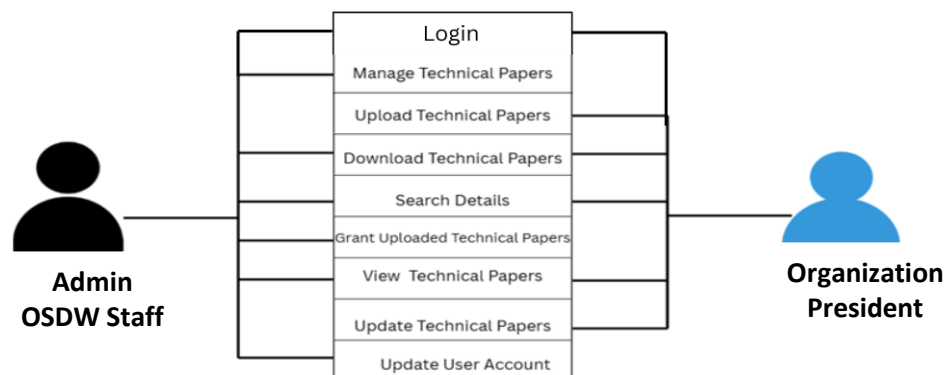
Figure 6. Context Diagram of the System



The context diagram illustrates the overall interaction of the CORMS (Campus Organization Report Management System) with its external entities. It shows how the Student Organization Officer submits reports and receives status updates, while the Administrator (OSDW Staff) manages users and reviews or approves submitted reports. The system serves as a centralized platform that processes all inputs and generates corresponding outputs, ensuring efficient monitoring and management of student organization reports.

To illustrate the interaction between users and the system, a Use Case Diagram was developed, as shown in Figure 7.

Figure 7. Use Case Diagram of the Proposed System





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Figure 7 presents the Use Case Diagram of the proposed system. The diagram shows the different users and their respective roles, privileges, and access levels. Each user type has specific permissions depending on their assigned role.

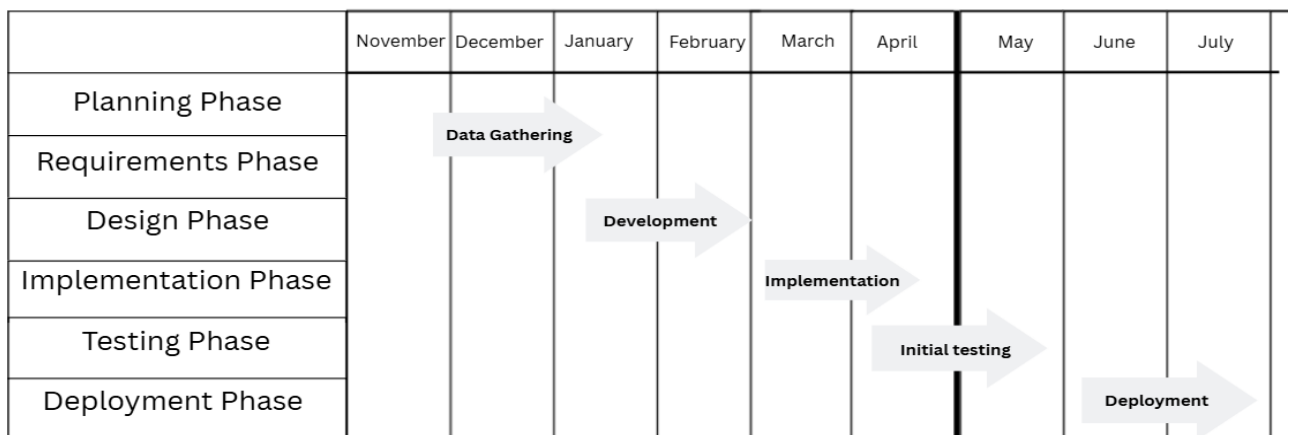
The major components of the system include organization information, activity details, narrative reports, accomplishment reports, summary reports, user registration, document uploads, and report viewing. The diagram demonstrates how these components interact within the system to support efficient information management.

Stage 4 – Implementation Phase

In this phase, the proponents implemented the system using appropriate development tools and technologies. The system was developed using Laravel as the web application framework and MySQL as the database management system. XAMPP was used as the local web server environment for testing, while CSS and Bootstrap were utilized for interface design and styling.

The implementation phase involved coding system modules, integrating database components, and configuring system functionalities based on the approved design specifications.

Figure 8. Implementation Phase





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Stage 5 – Testing Phase

Testing is a vital stage in the development of the proposed system. It involves executing the system with the objective of identifying errors, bugs, and performance issues. Functional testing was conducted to verify that each module operates according to its intended purpose.

Usability testing was also performed to ensure that the system is easy to use and efficient for its intended users. Identified issues were corrected before proceeding to deployment.

Stage 6 – Deployment Phase

After successful testing and necessary revisions, the system was deployed to the intended users. Evaluation was conducted using the software quality characteristics defined under ISO/IEC 25010 to determine whether the system satisfies operational and business requirements.

Feedback gathered from users provided insights into potential improvements and additional features. If adjustments or enhancements were requested, necessary revisions were made.

Stage 7: Maintenance

Maintenance may be required to keep it functional and up-to-date. As issues arise, proponents need to create patches and updates to address concern of the end users. If big issues may necessitate a return to the previous phase.

Locale of the Study

The study was conducted at Cagayan State University – Aparri Campus, particularly involving student organizations under the supervision of Office of Student Development and Welfare. The locale was selected because the campus currently utilizes manual or semi-manual documentation processes in managing and monitoring student organization reports, particularly in the collection and evaluation of mandatory submissions such as narrative, accomplishment, and summary reports,



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making it an appropriate environment for system implementation. Students Organization are required to prepare and submit various reports to the Office of Student Development and Welfare. This office is responsible for receiving, monitoring, and evaluating these submissions to ensure that all activities conducted by the organizational are properly documented and aligned with their GPOA.

Participants of the Study

Table 1 Participants of the Study

Participants	n	%
IT Experts	10	10
OSDW staff	5	5
Organization adviser	35	35
Student Leader	50	50
Total	100	100

The following are the participants of the study who contribute in the analysis, design, testing and implementation of the system, systems assessments, major and minor issues and concerns in the process of the "Design and Development of the CSORMS: A Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment and Summary Reports."

Data Gathering Tools or Instruments

The following are the instruments used in the collection and analysis of the required data.

Interview Guide - A structured interview guide was prepared to collect qualitative information regarding the current practices, policies, problems, and issues encountered in managing narrative, accomplishment, and summary reports of campus student organizations. The interview was



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conducted with student organization officers, organization advisers, and administrative staff from the Office of Student Development and Welfare. The responses gathered were used to identify system requirements and necessary features of the proposed system.

Questionnaire - The proponents used two (2) sets of questionnaires for the evaluation and assessment of the proposed project. The questionnaires are as follows: (a) ISO/IEC 25010 software quality standard and (b) Technology Acceptance Model, composed of different criteria and various interpretations that assess the extent of the proposed project in terms of software quality standards and user acceptability.

The proponents distributed the assessment questionnaires to the end-users and IT experts who evaluated the proposed project. The questionnaires were used to allow the participants to evaluate the system based on different criteria and corresponding interpretations according to their assessment and estimation of the proposed system's functionality, usability, performance, and overall acceptability.

Observation – the proponents obtain the different policies, guidelines, and standard operating procedures of the target agency through the respondents via observation. This allows the researchers to have a better understanding of the actual situation and status of the current processes and system in place.

Internet – this method is used to obtain in researching more needed information and incoming up with a better understanding pertaining to the different problems in the study.

Data Gathering Procedures

The data gathering process was conducted systematically to ensure accuracy and reliability of the collected information.



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First, the researchers secured formal approval from the campus administration to conduct the study at Cagayan State University. After approval, the researchers coordinated with the Office of Student Development and Welfare and recognized student organizations.

During the initial phase, interviews were conducted to gather information regarding the existing documentation and monitoring procedures. The identified problems and system requirements served as the foundation for the design and development of the proposed system.

After the development and testing of the system, the evaluation phase was conducted. The ISO/IEC 25010-based questionnaire was distributed to selected IT experts to assess the technical quality of the system. Meanwhile, the UTAUT2-based questionnaire was administered to student organization officers and administrative staff to determine the level of user acceptance.

The accomplished questionnaires were collected, tallied, and prepared for statistical analysis. The results were used to determine whether the developed system met the required software quality standards and achieved acceptable levels of user adoption.

Data Analysis

The section explains the different techniques in the analysis of the assessment or evaluation conducted. The use of weighted means, rank, percentage, and frequency counts as well tools to test significant differences in groups are widely seen. In this section, the Likert scale used in assessing software characteristics, usability, and performance.

Knowing the current practice, policies, problems and issues the current archiving system of the campus. In order to create the "CSORMS: A Web-Based Monitoring System for Campus Student Organization Narrative, Accomplishment and Summary Reports." for Cagayan State University-



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Aparri Campus, the proponents followed the appropriate rules of System Development Life Cycle (SDLC).

This serves as a guide to the proponents in their proposed system and its network infrastructure. System Planning, System Analysis, System Designs, Coding, Testing, Implementation, and System Maintenance are all phases of the SDLC paradigm.

Table 2. Likert's Scale by Rensis Likert

Weight	Interval	Verbal Interpretation
5	4.21 - 5.00	Strongly agree/ Very High Extent
4	3.41 -4.20	Agree/ High Extent
3	2.61 - 3.40	Neutral/ Moderate Extent
2	1.81 - 2.60	Disagree/ Low Extent
1	1.00 -1.80	Strongly Disagree/ Very Low Extent

Likert Scale - It functions by having a person complete a questionnaire that requires them to indicate the extent to which they agree or disagree with a series of statements.

Weighted Mean - The proponents is used the mean to measure the ratings of the respondents regarding on the performance and effectiveness of the proposed system.

The proponents used the mean to measure the ratings of the participants regarding the evaluation and assessment of the develop system. The formula in getting the mean is presented in the figure below:

$$\text{Weighted Mean (WM)} = \frac{\sum f(x_1 + x_2 + \dots + x_n)}{N}$$

Where:

$f(x_1 + x_2 + \dots + x_n)$ N= Represents the summation of all means of each criterion.



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N = number of criteria

Percentage was used to determine the qualitative ratio of the data gathered, also, is the number or ratio expressed as a fraction of 100. Percentage (P)= $FN \times 100$

Where:

- F = frequency of a numeric rating for some given criteria
- N = represents the total number of the participants.

Ethical Consideration

This study ensures that all ethical standards are properly observed throughout the research process. Participation of respondents, including student organization officers, advisers, and OSDW staff, is voluntary. They are informed about the purpose of the study and their consent is obtained before gathering any data. Confidentiality and privacy of the participants are strictly maintained. Any personal information collected is used solely for academic purposes and is not disclosed to unauthorized individuals. Responses are treated with anonymity to protect the identity of the participants. Furthermore, the study ensures that all data gathered from student organizations and the Office of Student Development and Welfare (OSDW) are handled responsibly. No information is manipulated or misrepresented. The researchers also ensure that the system developed does not harm, disadvantage, or violate the rights of any user. Lastly, proper acknowledgment is given to all sources used in the study to avoid plagiarism and to uphold academic integrity.



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